GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN IN PUBLICA COMMODA SEIT 1737

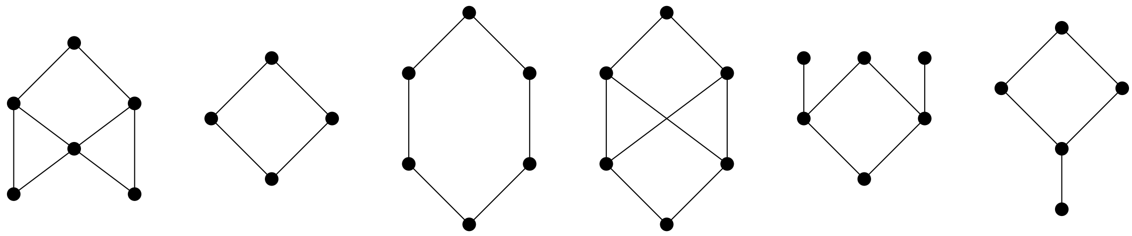
Compiler Construction (Winter 2025)

Exercise Sheet 7

Programming Languages and Compilers
Prof. Dr. Roland Leißa
Al-Harith Farhad, M.Sc

Exercise 7.1 Finite Lattices

1. Which of the partially ordered sets described by the following Hasse diagrams are lattices? Which of them are complete lattices?



2. Relate the set of all finite lattices with the set of all finite complete lattices. Prove your claims.

Exercise 7.2 Monotonicity and Expansiveness

In the lecture we saw the following two properties of a function $f : A \to A$ over some partially ordered set $(A, \sqsubseteq)$:

- Monotonicity: $\forall x, y \in A.\ x \sqsubseteq y \Rightarrow f(x) \sqsubseteq f(y)$
- Expansiveness: $\forall x \in A.\ x \sqsubseteq f(x)$

Show that, in general, neither implies the other.

Exercise 7.3 Chaotic Iteration

For computing the least fixed-point of the abstract semantics of a program, we have shown the correctness of the Kleene iteration scheme. Following this scheme, if we have an analysis that computes information for each program point, we have to apply the transformers for all program locations in parallel in each step (starting from $\bot^n$).

In the lecture, we claimed that we can instead apply transformers for individual program locations independently in an arbitrary order and still obtain the same least fixed-point as Kleene iteration. This exercise proves the validity of this claim.

First some definitions and notations:
Let $(L, \sqsubseteq)$ be a complete lattice with no infinite ascending chains. We extend the domain of $\sqsubseteq$ to n-tuples of elements of L in a straight-forward way: $\underline{a} \sqsubseteq \underline{b} \equiv \forall i \in \{1, \dots, n\}.\ a_i \sqsubseteq b_i$.

Let $F : L^n \to L^n$ with $F = (f_1, \dots, f_n)$. Let further all $f_i : L^n \to L$ be monotone. (From this also follows that F is monotone.) For convenience, we lift the component functions f_i to return n-tuples:

$$\widetilde{f_i} : L^n \to L^n,\ \widetilde{f_i}(\underline{x}) = (x_1, \dots, x_{i-1}, f_i(\underline{x}), x_{i+1}, \dots, x_n)$$

Note that the monotonicity of the $\widetilde{f_i}$ follows from the monotonicity of the f_i.

We add a shorthand notation for compositions of these functions: Let $\sigma \in \{1, \dots, n\}^*$ be a sequence of indices. We define:

$$\widetilde{f_\sigma} = \begin{cases} \lambda x.\ x & \text{if } \sigma = \varepsilon \\ \widetilde{f_k} \circ \widetilde{f_{\sigma'}} & \text{if } \sigma = k\sigma' \end{cases}$$

Proceed as follows to prove the correctness of the chaotic iteration scheme:

1. Prove that for any sequence ρ of indices, the intermediate results of $\widetilde{f_\rho}$ form an ascending chain:

$$\forall \sigma \in \{1, \dots, n\}^*.\ \forall k \in \{1, \dots, n\}.\ \widetilde{f_\sigma}(\bot^n) \sqsubseteq \widetilde{f_{k\sigma}}(\bot^n)$$

   Hint: Use an induction over the length of σ.

2. When performing our chaotic iteration, we apply component transformers $\widetilde{f_i}$ to $\perp^n$ until no application of any $\widetilde{f_i}$ can change the the result anymore. Argue that chaotic iteration terminates and computes a fixed-point of F.

3. Prove that the fixed-point x of F that we compute via chaotic iteration is the least fixed-point of F.
Hint: Compare the intermediate values of applying $\widetilde{f_\sigma}(\perp^n)$ *to those of applying* $F^{|\sigma|}(\perp^n)$.

Exercise 7.4 SSA-Property

Which of the following control flow graphs represent valid SSA-form programs? Justify your answer.